

Early Career Patterns: A Comparison of Great Britain and West Germany

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The transition from initial education to work has received a great amount of attention, but hardly any research treats this process holistically. This paper focuses on the serial succession of statuses instead of on single events in the early years after leaving full-time education. As a methodological tool sequence analysis will be applied. Optimal matching procedures allow for the direct comparison of entire career sequences taking into account the ordering of the events. The objective of applying this rather new tool is to empirically identify distinct patterns of transition into the labour market. The analysis covers the whole range of employment statuses including periods of unemployment and inactivity that individuals experience within the first five years following completion of education. A cross-national comparison between Great Britain and West Germany aims at investigating to what extent the observed patterns of transition processes are shaped by their institutional embeddedness.

Introduction

The transition from education to work has been the topic of much research. Most of this work, however, focuses on specific events, single time points, and individual transitions, not on the resulting trajectories and life courses. Little research has dealt with the transition and allocation process as sequences of several events. It can be assumed, however, that the allocation process in the labour market is not finished after reaching one's first occupational position but is instead a process of reaching a more or less stable position in the labour market. The aim of this paper, therefore, is to model the entry into work as a stepwise succession of different statuses by using sequence analysis. Sequence analysis allows us to handle the whole career history by taking into account the information about length and ordering of different statuses instead of single events or time points. That this is not just a methodological issue but also of substantial importance becomes clear if we think of the entry into work as a process evolving over time

which can only be captured by concentrating on information about whole trajectories. Thus, as an innovative methodological tool, we suppose that this method will provide new insights into the dynamics and distribution of early career trajectories. Furthermore, it also enables us to tackle the problem of developing typologies of trajectories in a promising way. Previous research dealing with typologies of transition patterns basically followed a more qualitative approach restricted to small sample sizes. With the statistical analysis of the whole career history information for representative datasets, we can now put the results on a much broader basis. This is especially necessary for an international comparison of different transition patterns.

It is a well-known fact that allocation and matching processes differ widely between different structural and institutional contexts. An international comparison is therefore helpful for better understanding how the institutional context people

live in shapes their entry into work. Investigation of Great Britain and West Germany (in the following referred to as 'Germany') offers a variation in relevant dimensions of the educational system and labour market structures. Previous research has shown the importance of several dimensions influencing transition and entry patterns and the allocation to occupational positions. Among these are characteristics of the education and training system, the labour market structure, and the structuring and regulation of the transition process itself. Nations vary across relevant dimensions of these systems and with regard to the linking of these systems (Kerckhoff, 2001; Shavit and Müller, 1998; Hannan, Raffe, and Smyth, 1997; Marsden, 1990; Allmendinger, 1989; Maurice, Sellier, and Silvestre, 1986; Doeringer and Piore, 1971), so that we expect young people typically to follow different entry patterns into work. These entry patterns vary with regard to several dimensions, e.g. their achievement of specific statuses and positions, the mobility between these – what we could call 'early career turbulences' (Kerckhoff, 2001) – the time it takes to achieve a stable position, as well as the age at entry and the amount of individual choices to be made. This list is of course far from being complete.

The main focus of this paper is on the question whether distinct patterns of entering the labour market can be empirically identified and how they differ between different institutional contexts. More precisely, it will be investigated to what extent equivalent or different entry patterns into work are identified in different countries, namely Great Britain and West Germany,¹ and how these patterns differ with regard to their frequency and composition between the two nations. Differences between gender and the time period are taken into account. As the emphasis is on the process of entry into work, this paper concentrates on the first five years after leaving full-time education drawing on monthly employment status information.

The remainder of the paper will proceed as follows: the next section provides a brief overview of the different institutional contexts within Germany and Great Britain and their expected impacts on the resulting trajectories. There is then an introduction to the applied method, sequence analysis, and some details about the data and the sample selection this paper draws on. Empirical

results are the subject of the next section: after a brief description of the early career sequences in general, the analysis focuses on the investigation of typical career patterns. The paper concludes with a summary and discusses the empirical findings in the light of the outlined hypotheses, followed by some more general remarks about the new insights this approach is able to provide.

Theoretical Considerations

In almost all western countries we have observed prolonged transitions to work (in terms of the time when young people first leave the education and training system) over recent decades. Despite this general finding, Germany is among those with the lengthiest processes. Among other reasons this is due to the strong emphasis on educational credentials, leading to a high participation rate in education after compulsory schooling. This is true especially for participation in vocational training, mostly within the dual system of apprenticeship. Through this specific training the transition period in Germany is highly structured and regulated (Allmendinger and Hinz, 1997). In contrast, in the UK educational credentials do not carry the same weight either for young people themselves or for the employers in their recruitment decisions. Young people enter the labour market much earlier and through more 'informal' channels. Transition to work takes place earlier in Britain and without the structural regulation observed in Germany.

Dimensions of the education and training system are of central importance for the link between educational achievements and labour market outcomes, as well as for the way this transition between the two systems operates. As relevant dimensions of the education and training system the extent of standardization and stratification and their horizontal differentiation are usually noted (Allmendinger, 1989; Kerckhoff, 1996; Shavit and Müller, 1998; Kerckhoff, 2001; Shavit and Müller, 2001). Among other things these dimensions shape the connection with the labour market, as they influence the way labour markets perform and employers select potential employees. The German educational system is typically classified as highly standardized and stratified, whereas the education and training

system in the UK shows a moderate degree of differentiation with high standardization only within general education (Hannan, Raffe, and Smyth, 1997; Kerckhoff, 2001). This results in a strong linkage between educational achievement and labour market outcomes in Germany, whereas labour market position is not determined to the same extent by education in the UK.

During recent decades the British education and training system has undergone a lot of change (Brauns and Steinmann, 1999; CEDEFOP, 1994; Oulton and Steedman, 1995; Roberts, 1995; Lindley, 1996); vocational training has been subject to substantial reform and new routes into employment have been established (Bynner and Roberts, 1991; Evans and Heinz, 1994).² In contrast, the German system is characterized by relative stability.

Closely connected to the characteristics of the education and training system is the segmentation of the labour market. The German system has promoted the emergency of an occupationally segmented labour market (Marsden, 1990; Marsden and Ryan, 1995; Müller and Shavit, 1998). Standardized vocational training is an important factor for the development of such a market. The literature is not clear in the categorization of the UK, but recent studies indicate that the UK is closer to the ideal type of an internal labour market.³ This implies that young people enter mainly as unskilled and training is to a larger degree provided on the job, including a period of screening the workers due to the lack of reliable signals. Employees are provided with specific skills depending on their employers' needs, and these are not automatically transferable to other organizations. These labour market structures have strong implications for the mobility patterns people follow in their early career. Internal markets lead to a much higher mobility in the early years of labour force participation, while within occupational markets relatively stable positions are gained more easily, but without considerable prospects for upward mobility later.

Due to different institutional frameworks in Germany and the UK the typical patterns young people follow in entering the labour market are supposed to vary considerably. Although both countries have experienced some common developments, the impact of these has differed in degree. Among them are educational expansion, growing flexibility of the

labour market, rising unemployment rates and therefore an overall prolonged transition into work,⁴ both in terms of the age of first entering the labour market and the time needed to establish oneself within it (Roberts, Clark, and Wallace, 1994; Roberts, 1995). The highly standardized nature of the education and training system in Germany and its vocational specificity, combined with a high degree of regulation of the transition process, especially within vocational training at the secondary level, leads to smoother and more standardized transitions into work. These factors all facilitate the process of matching jobs and workers, which leads to lower turbulence at labour market entry. Through standardized education, employers receive reliable signals (Spence, 1981). Information about the job relevance of skills is also available, as many young people enter the work force as skilled workers, already provided with training in a specific occupation. Moreover, through work-based training a first screening of potential employees is possible. Therefore mobility at labour market entry is supposed to be low.

In contrast, for the UK we expect less standardized and smooth transitions into work, characterized by higher mobility at labour market entry. Mobility occurs as entry into and exit from employment in terms of a faster rate of job change as well as a faster upward mobility, for those with successful entry. Periods of unemployment are common among entrants. The matching of jobs and workers has to be elaborated individually, as many young people are entering with no training in a specific occupation. Due to the lack of reliable and job-relevant screening information, the screening process is in some sense extended to the first period on the job. As a consequence greater differences between the youth and adult labour markets are observed in the UK.

Entry patterns are expected to have undergone more change over time in the UK. This is due to the reform of the education and training system but also to the greater flexibilization of the labour market. Although we find trends towards more flexible markets even in Germany, flexibility in the UK is considerably higher. Flexibilization implies that 'non-standard' working relations are becoming more important in terms of frequency; continuous full-time work is becoming less frequent, and at

the same time part-time work, limited contracts, and unstable and precarious employment are gaining in importance (Standing, 1995; Rodgers, 1989). These trends are supposed to be most pronounced at career entry, leading to a declining proportion of direct and stable entries into work.

More specifically, we therefore expect for Germany a higher proportion of people entering the labour market directly after leaving the educational system through access to more or less stable full-time employment and remaining in this status. Although unemployment in general also became more common in Germany, patterns showing considerable spells of unemployment are supposed to be more frequent in the UK and to have increased over the last decade. Unemployment in general is a signal for problematic and unstable entry, but with the chosen focus a more differentiated picture can be provided through further differentiation of these career patterns. Patterns showing different periods spent in unemployment and also of varying location and order can be distinguished. (These patterns are supposed to reveal those with only short 'search-unemployment' experience at career entry and patterns of long-term unemployment as well as patterns with multiple spells of unemployment, characterized by greater instability.) Adopting a broader view, unstable and 'chaotic' patterns should be more frequent in the UK and should have gained in importance across the entry cohorts. The share of early career patterns showing spells of self-employment are, for reasons of greater labour market flexibilization, also supposed to be more frequent in the UK.

Methodology and Data

'We are accustomed to view the cross-sectional structure of employment in a society... as the major determinant of life-chances and the class structure... But considered more carefully, it emerges that the number of people who happen to possess particular occupational characteristics at one historical point has in fact no necessary implications for the pattern of stratification in a society. What is important is not the number in each occupation at any one time, but rather the individuals' *trajectories through various occupations over time*.' (Gershuny, 1993:

136, emphasis in the original). Thus, on the basis of cross-sectional distribution, it is not possible to depict the mobility and thus the mechanism leading to distributions. The focus therefore has to be on pathways, on trajectories rather than on distribution, to overcome this static and sometimes misleading approach.

Most of the 'classical' methodological tools used for analysing longitudinal data focus on single events, instead of mobility patterns in terms of a serial succession of sequential events. In contrast, sequence analysis considers whole career sequences (as careers can be represented as a serial succession of different statuses over time) and therefore makes it possible to treat these data holistically. The handling of whole sequences is exactly what existing analyses of career mobility leave out (see Chan, 1995; Brückner and Rohwer, 1996; Erzberger and Prein, 1997; Halpin, 1996; Halpin and Chan, 1998; Rohwer and Trappe, 1997 for counter-examples). This approach is able to provide viable answers concerning typical patterns in unique sequences; it allows one to empirically identify typologies of sequences. The specific technique used is Optimal Matching Analysis (henceforth: OMA; Rohwer, 1996, Abbott, 1995, Dijkstra and Târis, 1995), a technique introduced by molecular biologists investigating e.g. DNA sequences (Sankoff and Kruskal, 1983). It was first applied to sociological analysis by Abbott (Abbott and Forrest, 1986; Abbott and Hrycak, 1990). However, OMA by itself does not give any classification, but directly creates interval-level measures of dissimilarity between sequences which can be used as input to further analysis such as cluster analysis or multi-dimensional scaling. To investigate typical early career patterns and their comparison over different countries, after the first step of comparing the sequences by calculating their resemblance by OMA, a second step is needed to bundle up groups based on their similarity.

The basic idea underlying the OMA algorithm is a simple count-metric. What OMA does is count for the 'costs' needed to turn sequence A into sequence B or vice versa by counting the minimum number of transformations needed to make sequence A equal B. The more steps that have to be taken to make the sequences equal, the more costs occur and consequently, the greater is their dissimilarity. Two types of transformations are possible: substitutions

and insertions/deletions.⁵ With this procedure OMA takes not only the length and frequency of the events into consideration, but also their location and order.

In principle there are two options for calculating these differences. Dissimilarities can be calculated either pairwise by comparing each sequence with each other or – as the second alternative – in comparison to a specified reference sequence. I use both options for computing the distances. First the sequences are compared with the standard reference of continuous full-time work, and then the distances of pairwise comparison as the basis for the clustering procedure are reported.

As an additional feature to bring theoretical considerations into the calculation of the distances, transformations can be weighted. It would be far beyond the purpose of this paper to cover all relevant aspects of weighting, but a few remarks should be made. First, different substitutions can be assigned different weights. One can think of some statuses as being closer together than others and therefore assign different costs for their substitution. A special ranking of the statuses has to be decided theoretically. The first set of results reported are based on such weighting; some additional remarks are given in the Appendix, which also shows the matrix of the substitution costs that have been applied. Secondly, weighting can also be assigned to insertion and deletion costs. These two transformations are treated in the same way – i.e. are assigned the same costs – and are often briefly described as ‘indel’-transformation. It seems rather obvious that the insertion or deletion of a specific status can cost more than others if it is quite unusual. But these costs are in some sense a function of what the sequence already looks like: for one sequence the occurrence of a specific status might be rather unrealistic, while for another it might not. To decide this point one has to make so many assumptions that I have decided to treat any insertion and deletion as having the same cost as any other, following other authors. It has to be proved whether these cost-assignments produce meaningful categorizations. To complicate things further, it could also be important to take into account the point in time at which specific transformations are carried out and therefore assign different weights to the same transformation at different points of time within

the career sequence. This is where we reach the limits of this approach for the moment. The temporal order in that sense simply cannot be taken into consideration when calculating the matching costs.

As mentioned above, optimal matching does not provide any classification of the sequences. Therefore a second step to investigate for typical groups of entry patterns is needed. Classification can be reached in many ways and weight can be put on either theoretical considerations or the empirical investigation. In this paper the latter approach was chosen and classification of the groups is obtained empirically drawing on cluster analysis. Precisely, the pairwise dissimilarity measures are subjected to cluster analysis. Clustering data by cluster analysis is always a problematic procedure, as it provides a cluster solution even if there is no meaningful structure within the data,⁶ and different rules of cluster formation lead to different solutions for the same dataset. Therefore one of the critical decisions to make is the choice of the clustering algorithm. Every algorithm has its own advantages and problems. I decided to choose Ward, which is supposed to find the most homogeneous clusters⁷ and seems to provide the most reasonable result for this analysis. Cluster analyses are conducted separately for each of the two countries.

Empirical analyses draw on data (concentrating on Great Britain and not the entire UK) from the British Household Panel Survey (BHPS)⁸ and the German Socio-Economic Panel (SOEP), covering an interval from 1985 to 1996. For Germany we draw just on the West German population. To investigate the entry process into the labour market, the first five years after leaving the full-time education and training system (ETS) are covered, which implies that the latest entry took place in 1991. This selection results in 1111 observations for Germany and 1114 for Great Britain. A brief description of further modifications of the datasets is given in the Appendix.⁹

Apprenticeship training is considered part of the educational system. Individuals thus enter the sample after their periods within apprenticeship.¹⁰ Due to the lack of standardized training content the British government training schemes (GTS) are more or less understood as active labour market policies rather than real education and thus

considered part of the labour market experience and consequently included in the early career sequence. In addition military service is excluded from analysis in Germany. As the main purpose is to investigate labour market entry process this seems reasonably justified.¹¹

The definition of the early career sequences is based on monthly information on employment status. Concentration on employment status has the advantage of covering the whole range of statuses, including spells of unemployment and non-workforce participation.¹² The following status categories have been covered:

- Self-employment;
- Full-time dependent employment;
- Part-time dependent employment;
- Government training schemes (GB only);¹³
- Unemployment;
- Full-time education (return to);
- Family care/maternity leave; and
- a heterogeneous ‘other’ category including a group with status not specified further, military service (in GB only – see appendix for details) and retirement.

Government training schemes have no counterpart in Germany. We therefore end up with one additional status in Britain.¹⁴

With the definition of ‘ETS leaving cohorts’ different age groups are compared, depending on their age at leaving education, mirroring to a large extent their educational level within different time periods, but at equivalent stages in their career history. For both, entry period and age can be controlled for, and for both, effects on the performance at labour market entry are expected. The entry date is taken into account in a dichotomous way: the ‘old-entry cohorts’ refer to entry cohorts of 1984–1988 and the ‘young-entry cohorts’ refer to those entering between 1989 and 1991.¹⁵

Education is measured by the CASMIN scale, which was developed especially for comparative research (see Brauns and Steinmann, 1999 for a detailed description; Koenig, Luettinger, and Mueller, 1988 for the original version).¹⁶ CASMIN differentiates between the level of education and general and vocational tracks. It therefore allows for the representation of non-linear impacts of

education. A short description is given in Table A1 in the appendix.

Empirical Results¹⁷

Deviation of the Early Career Patterns from a Standard Sequence of Continuous Full-Time Work

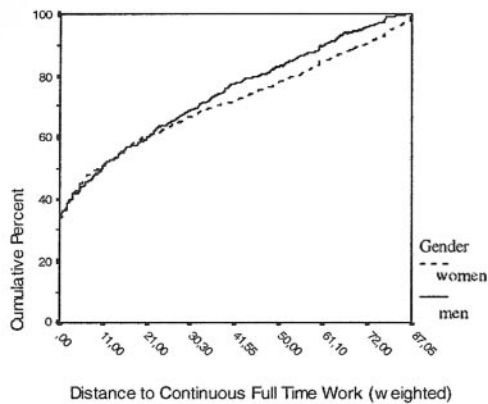
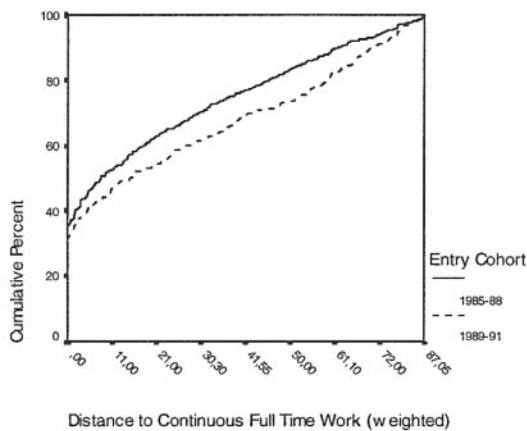
In order to give a first impression of early careers in the two nations, individual sequences are first compared with a standard sequence consisting of continuous full-time work. Table 1 contains the proportion of people working continuously full-time over the first five years. For a more differentiated picture figures 1a, 1b, 2a, and 2b show the cumulated frequencies of distances to continuous full-time work subdivided by gender and entry cohorts. Results are based on weighted distance calculation (see Appendix) but no substantial differences occur using non-weighted results.

The largest proportion of people, 34.0 per cent in Germany and 33.0 per cent in Great Britain, are continuously within full-time dependent work during their first five years after leaving the education and training system. This proportion clearly differs between the two sexes and between the two entry cohorts under investigation, though to varying degrees in the two countries (see Table 1). Gender differences are very pronounced in Germany, with males clearly dominating the pattern of continuous full-time work. But a closer inspection reveals gender differences concerning this proportion in GB also: while women of the older entry cohorts even exceeded the portion of males remaining in full-time work in the first five years, they experienced a decline of more than 10

Table 1. *Proportions in continuous full-time work over the first five years (%)*

	Great Britain		Germany	
	Men	Women	Men	Women
Entry cohort 1985–88	34.4	36.2	41.5	26.1
Entry cohort 1989–91	30.6	24.4	40.7	24.1
Overall	33.0		34.0	

Source: BHPS and SOEP, labour market entrants (first 5 years after leaving the ETS).

1a by Gender, Great Britain**1b by Gender, Germany****2a by Entry Cohort, Great Britain****2b by Entry Cohort, Germany**

Sources: BHPS and SOEP. Cumulated distances to continuous full time work (employment status information). Distance calculation by optimal matching. Labour market entrants (first 5 years after leaving the ETS). All figures based on a weighting of the substitution costs

Figure 1a, 1b, and 2a, and 2b. *Distance to continuous full-time work, cumulated frequencies*

percentage points in this pattern over the two entry cohorts.¹⁸ Although we observe declining full-time work participation between the second half of the 1980s and the beginning of the early 1990s in all groups, percentages in Germany stay more or less stable over the years.

In addition to this proportion in continuous full-time work, the more detailed inspection of distances to these career patterns provided in the figures (1a to 2b) also shows substantial differences between the two countries.¹⁹ They clearly support the hypotheses concerning more pronounced gender differences in Germany and considerably more change over time in Great Britain.

For Germany the 'gender gap' is obvious: females do not follow the continuous full-time work pattern to the same extent as men and higher proportions have higher distances to this standard reference. Although women in GB are also significantly further away from male patterns with regard to non-zero distances, gender differences there are rather weak. Beside the well-known fact of higher male full-time work participation especially in Germany, this confirms that women's early career patterns also diverge from this norm. To investigate whether not only the participation rates in several statuses, but also the entry patterns into work differ, I concentrate on those who have actually entered the active work force after a certain period of time, e.g. after five years: only those in full-time dependent work after this time are considered, so as not to confuse the different processes governing entry patterns and participation decisions.²⁰ Results (not reported here) show that for Germany these entry patterns into full-time work differ significantly between the two genders, with males starting much earlier with full-time work and a tendency to remain in it. In GB, however, gender differences again are not apparent at first sight: significant differences occur only if we disregard those who directly enter this status and remain within it (e.g. those within continuous full-time work). So in addition to different participation decisions, we also find different entry patterns into work, but to a considerably lower degree in GB.

Focusing on differences between the two entry cohorts, we can see a shift away from this standard over time for GB, indicating greater change in the

British labour market. With regard to these distances Germany is characterized by high stability: no change between the two entry cohorts can be found. However it should be added that with this distinction of entry cohorts it is not possible to grasp change over time satisfactorily; for this purpose a broader time period would have had to be covered.

Deeper insight into the role educational achievements play in influencing this deviation from the chosen standard sequence is obtained by a regression analysis of the distances. This is possible, as one of OMA's advantages lies in producing interval-level distances.

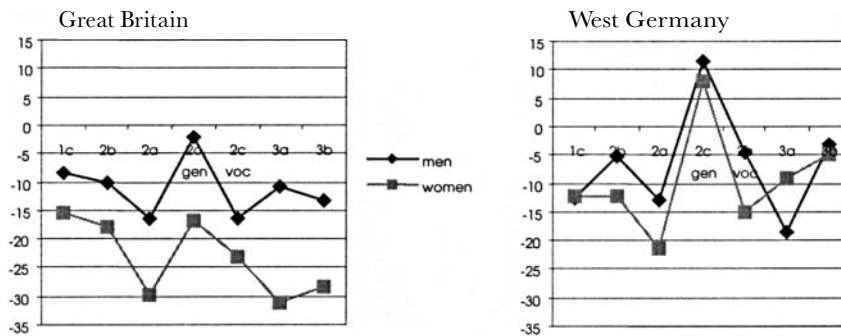
Results in Figure 3 refer to the estimated model reported in Table 2. Table 2 repeats the gender differences in distance to continuous full-time work. Intercepts for women also indicate that when controlling for educational achievement men are much closer to the labour market in both countries. Direct comparison of the values between the two countries is not recommended, however, given the one additional status in GB. This approach shows almost no differences between the entry cohorts. With regard to the pattern of how educational achievements influence the distance to continuous full-time work, results show similarities as well as dissimilarities across the countries. Education in general reduces the deviation from continuous full-time employment. With the exception of full general secondary level (CASMIN 2cgen) for Germany, education increases the probability of being closer to continuous full-time work for both sexes; but having only the *Abitur* in West Germany leads – at least for men – to a career with even more distance to work than possession of only basic qualification. It is striking that the crucial dimension which provides access to rather stable early careers is the possession of vocational training and not the level of educational achievement. In principle this applies for both countries. In Germany it is especially apparent for those who obtained apprenticeship training, supporting the hypothesis about the importance of work-based training, whereas in GB the level of educational achievements seems slightly more important (indicated by the more linear pattern of influence and the higher advantage of tertiary-level education compared to Germany).

Table 2. OLS regression analysis of the distance to continuous full-time work, *b* coefficients

	Great Britain		Germany	
	Men	Women	Men	Women
Additional descriptive measurements				
Range of distance	0–85.50	0–87.05	0–90.00	0–90.00
Mean distance	21.78	24.21	14.63	29.12
Standard deviation	24.55	28.19	21.83	29.25
Entry cohort 1989–91 (Ref.: 1985–88)	4.93	6.19*	0.24	1.24
CASMIN (Ref.: 1ab social minimum educ.)				
CASMIN 1c (basic voc. training)	–8.27	–15.50**	–12.64**	–12.17**
CASMIN 2b (intermediate sec. level, gen. educ.)	–10.21**	–17.88**	–5.16	–12.42*
CASMIN 2a (intermediate sec. level, voc. educ.)	–16.62**	–29.88**	–12.95**	–21.46**
CASMIN 2c gen. (higher sec. level, gen. educ.)	–2.25	–16.89**	11.33*	8.03
CASMIN 2c voc. (higher sec. level, voc. educ.)	–16.42**	–23.20**	–4.72	–15.13**
CASMIN 3a (lower level tertiary)	–10.89**	–30.98**	–18.43**	–8.99
CASMIN 3b (higher level tertiary)	–13.23**	–28.52**	–3.23	–4.90
Intercept	29.74	42.84	21.95	39.73
R ²	7.5%	14.1%	10.1%	9.1%

**significant at the 1% level; * significant at the 5% level.

Source: BHPS and SOEP, labour market entrants (first 5 years after leaving the ETS).



Source: BHPS and G-SOEP, labour market entrants (first 5 years after leaving the ETS)

Figure 3. Educational effects in reducing the distance to continuous full-time work

Measurement of education, however, is not optimal as it is based on information at one fixed time point (1991). This implies, at least for Great Britain, that older cohorts have a higher chance of reaching additional qualifications which we are not able to control for in this approach, given the relatively high proportion attending (part-time) further education. To take into account this short-

coming of educational measurement at least to some extent, those individuals returning to education as their first status within these five years were excluded from analysis in an additional step. The overall effect of education becomes slightly more pronounced but no fundamental changes are apparent. Of special interest is the fact that the specific pattern of general secondary education (2c general) for Germany and

Table 3. *Description of the typical entry patterns (% and N)*

Typical entry pattern: the clusters	Great Britain			Germany		
	%	Cum. %	N	%	Cum. %	N
'Full-Time Work'	51.7		566	53.4		591
'Full-time work plus other statuses'	11.3		124	12.3		136
		63.0			65.7	
'GTS (short) → full-time work'	4.7		51	—		
'GTS → full-time work'	5.2		57	—		
		9.9				
'Self-employment'	3.1		34	2.3		25
'Full-time work → self-employment'	—			2.2		24
		3.1			4.5	
'Part-time work'	2.9		32	3.5		39
		2.9			3.5	
'Unemployment—short'	3.2		35	2.0		22
'Unemployment—long term'	5.8		64	2.1		23
		9.0			4.1	
'Return to education'	3.4		37	2.4		27
'Return to education (later)'	—			5.1		56
		3.4			7.5	
'Work → family care'	2.8		31	7.0		78
'Family care'	3.7		40	6.0		66
		6.5			13.0	
'Rest/no clear structure'	2.2		24	—		
'Rest'	—			1.8		20
		2.2			1.8	
Total	100		1095	100		1107

Source: BHPS and SOEP, labour market entrants (first 5 years after leaving the ETS), results of the cluster analysis.

the relatively low pay-off of tertiary education remains stable: those findings are independent of returning to education later in one's career.

The following section investigates these early career sequences in greater detail.

Early Career Patterns: Description of the Clusters

For a closer look at what is behind this 'deviation' from continuous full-time work, I now turn to the investigation of the early career patterns. Can we identify distinct entry patterns empirically and how do they differ between the two countries?

The following results are based on pairwise distance calculation by OMA with no weighting. Although it is immediately reasonable to think of

some statuses being closer together than others, in cluster analysis groups turn out to be more clearly structured when no weighting is used. With the choice of non-weighted results emphasis is given to the ordering of the statuses, not only to their occurrence and duration. For Great Britain and Germany a twelve-cluster solution seems to provide the most reasonable grouping of the data.²¹ Cluster analysis was conducted separately for the two countries. The primary aim was to investigate whether and to what extent we find the same or distinct patterns emerging in the two countries. With cluster procedures strongly depending on input of distances, separate analysis is supposed to give more clearly structured results. However, it also generates a problem: how to decide that two separately emerging groups are equivalent or not. But, as we shall see later, patterns are quite straightforward.²²

Table 3 provides an initial description of the emerging patterns. The second column (Cum. %) in both panels for the two countries provides cumulated percentages of persons belonging to groups dominated by a specific status or showing a characteristic pattern. For instance, 9 per cent of the sample in GB and 4.1 per cent in Germany are significantly affected by unemployment. A more detailed picture provided by the emerging patterns shows that 5.8 per cent in Britain (64 persons) and 2.1 per cent in Germany (23 persons) are affected by ongoing labour market exclusion, those in long-term unemployment. With this presentation it is easy to obtain a first impression about the relevance of different early career patterns in the two economies. Table 4 shows the mean duration (in months)²³ within the seven (FRG) or eight (GB) statuses for the different typical career patterns. Figures in bold indicate the dominance of one (or two) statuses within the specific cluster. Out of the description of the clusters with regard to the duration spent in several statuses, insufficient or no conclusions can be drawn about the serial succession of the different statuses. Therefore the detailed picture of the early career sequences is given in Figures 4a and 4b. The figures report the whole career sequences. For each individual in the sample one line is drawn representing the status each month within the first five years. Status information is indicated by the different shading. Individual sequences are sorted by their cluster membership; black lines mark the cluster borders. The special advantage of this graph is that it allows for the detailed visual exploration of the typical trajectories. Besides the potential distinctions in the early career patterns that young people in the two countries typically follow, the emerging groups may also differ in their composition. Table 5 gives some details about the composition of the patterns with regard to gender, entry cohort, and the predominance of educational achievements. Due to the very few cases included in some of the emerging clusters, education is only considered in a rather rough categorization: the level of education and the possession of a vocational qualification are both treated dichotomously. Both categorizations are based on CASMIN classification. Those with higher than intermediate-level education are considered as having higher education; vocational qualification is considered without regard to the level.²⁴ In principle

the composition of the clusters with regard to educational achievement is quite similar in both countries, though with some significant deviations.

Probably the most striking result of these first descriptions is that these methods do produce meaningful and well-defined results, as for instance can be seen in Table 4, where clusters clearly differ in distribution over the several statuses. But as becomes clear from the detailed representation in Figures 4a and 4b and Table 5, patterns also differ clearly in their location and the order of statuses as well as in their composition.

For both countries the predominant pattern containing about 65 per cent of the sample is one mainly composed of people continuously working full-time, but this group is further subdivided into two groups, one consisting of more or less pure full-time working careers, the other showing additional spells in other statuses, mainly in unemployment. The first, homogeneous group, containing about 50 per cent of the sample, enters the labour market directly after leaving the education and training system through periods of full-time dependent work and remains within this status over their first years. On average they spend about 97 per cent of their first five years within full-time work (see Table 4), but as can be seen from the detailed report of sequences in Figures 4a and 4b, the vast majority have no additional spells in other statuses. The dominance of this pattern as well as the strong gender differences occurring for Germany with regard to this group were already outlined in the section above. Continuous full-time work participation is much more common among men than women in Germany, while there are no gender differences at first sight in GB. Behind this rather stable pattern, however, a lot of mobility can be hidden: this view takes into account neither employer nor job changes.

The second mainly full-time work-dominated group is characterized by much higher heterogeneity than the first one. In principle we detect this pattern in both countries but with considerably higher heterogeneity in GB. Besides the predominance of periods in full-time work (mostly) there is one additional spell of limited duration in another status, usually unemployment. These additional spells have no clear structure in location. In a sense these patterns can be interpreted as showing problems of entering work, as they are characterized

Table 4 *Time spent in the different statuses by cluster membership (months)*

Cluster	Self-empl.	FT work	PT work	GTS	Unem- ployed	Educa- tion	Family care	Other	N
Great Britain									
'Full-Time Work'	0.05	58.20	0.20	0.16	0.97	0.07	0.00	0.28	566
'Full-time work+other statuses'	1.44	39.98	3.35	0.81	7.31	4.53	1.62	0.86	124
'GTS (short) → full-time work'	0.22	46.31	0.09	11.51	1.20	0.07	0.33	0.25	51
'GTS → full-time work'	0.07	24.95	1.95	25.37	4.68	1.00	1.65	0.32	57
'Self-employment'	35.59	14.00	0.38	2.12	1.91	0.18	1.56	0.26	34
'Full-time work → self-employment'									
'Part-time work'	1.31	9.91	41.75	0.13	4.81	0.44	0.88	0.78	32
'Unemployment—short'	0.08	26.40	1.14	0.26	30.89	0.46	0.66	0.11	35
'Unemployment—long term'	1.63	6.09	1.52	3.03	46.08	1.16	0.00	0.48	64
'Return to education'	0.43	12.78	1.95	0.22	2.97	39.59	0.95	1.11	37
'Return to education (later)'									
'Work → family care'	0.00	24.61	2.97	0.35	1.55	0.29	30.23	0.00	31
'Family care'	0.00	1.78	3.42	3.28	7.20	0.15	43.85	0.32	40
'Rest/no clear structure'	1.33	19.42	2.33	2.33	9.25	10.21	0.00	15.13	24
'Rest'									
Total	1.62	41.61	2.27	2.48	6.12	2.26	2.91	0.72	1095
Germany									
'Full-time work'	0.02	58.72	0.08		0.54	0.35	0.14	0.15	591
'Full-time work+other statuses'	0.93	45.87	1.82		6.66	2.63	0.81	1.27	136
'Self-employment'	49.44	5.68	2.92		0.24	0.00	1.24	0.48	25
'Full-time work → self-employment'	19.79	35.08	0.71		1.21	1.62	10.8	0.50	24
'Part-time work'	0.92	7.05	38.37		4.08	0.92	7.87	0.28	39
'Unemployment—short'	0.00	29.91	1.14		25.73	1.77	0.55	0.91	22
'Unemployment—long term'	0.00	6.70	1.91		38.57	2.00	8.13	2.70	23
'Return to education'	0.22	10.85	3.41		6.37	37.89	1.07	0.19	27
'Return to education (later)'	0.00	30.23	0.11		2.38	25.39	1.34	0.55	56
'Work → family care'	0.42	34.81	4.95		2.55	4.32	11.73	1.22	78
'Family care'	0.09	7.76	2.32		4.53	0.53	44.52	0.26	66
'Rest/no clear structure'									
'Rest'	0.00	4.45	1.2		5.20	10.65	1.60	36.90	20
Total	1.74	43.64	2.38		3.42	3.39	4.29	1.14	1107

Note: Total time under investigation: 60 months. Bold numbers indicate dominant statuses.

Source: BHPS and SOEP, labour market entrants (first 5 years after leaving the ETS), description of the clusters.

by remarkable mobility in and out of employment. But for both countries rather different types are bundled together in this cluster. They range from (multiple) spells of education though periods in part-time work participation, to those with initial unemployment experience at labour market entry of rather short duration and multiple spells of unemployment after their first work experience.

Two additional clusters consisting of those starting with a period in government training schemes (GTS), such as Youth Training or the Youth Training Scheme, and successful transition into full-time work afterwards, emerge for GB. Obviously they can have no counterpart in Germany. Together they cover about 10 per cent of the sample. Both groups differ concerning their time spent within GTS: the

Table 5 *Composition of the 'typical patterns': proportion of males, entrants between 1989 and 1991, higher educated and vocationally trained in deviation from overall proportion (%)*

	Great Britain				Germany			
	Men	1989–91 ^a	High educ.	Voc. train.	Men	1989–91 ^a	High educ.	Voc. train.
Overall proportion	48%	31%	57%	58%	54%	41%	48%	67%
Entry pattern								
'Full-time work'	+1	−3	+11	+8	+11	+0	+4	+7
'Full-time work plus other statuses'	+0	−5	−2	−3	+5	+0	−7	−1
'GTS (short) → full-time work'	+7	−9	−12	+1				
'GTS → full-time work'	+8	+2	−15	+9				
'Self-employment'	+14	−13	−1	−5	+18	+3	+31	+1
'Full-time work → self-employment'					+21	+22	+9	+16
'Part-time work'	+5	+32	+2	−20	−21	−3	−5	+0
'Unemployment—short'	+23	+15	−31	−32	+23	−27	−31	−22
'Unemployment—long term'	+13	+19	−28	−23	−24	+7	−40	−41
'Return to education'	+1	+14	+24	−7	+9	+7	+3	−15
'Return to education (later)'					+9	+11	+22	−3
'Work → family care'	−48	−12	−22	−10	−41	−5	−3	−11
'Family care'	−48	+2	−42	−35	−54	−3	−17	−22
'Rest → no clear structure'	+10	−6	+10	−4				
'Rest'					+1	+21	−29	−47

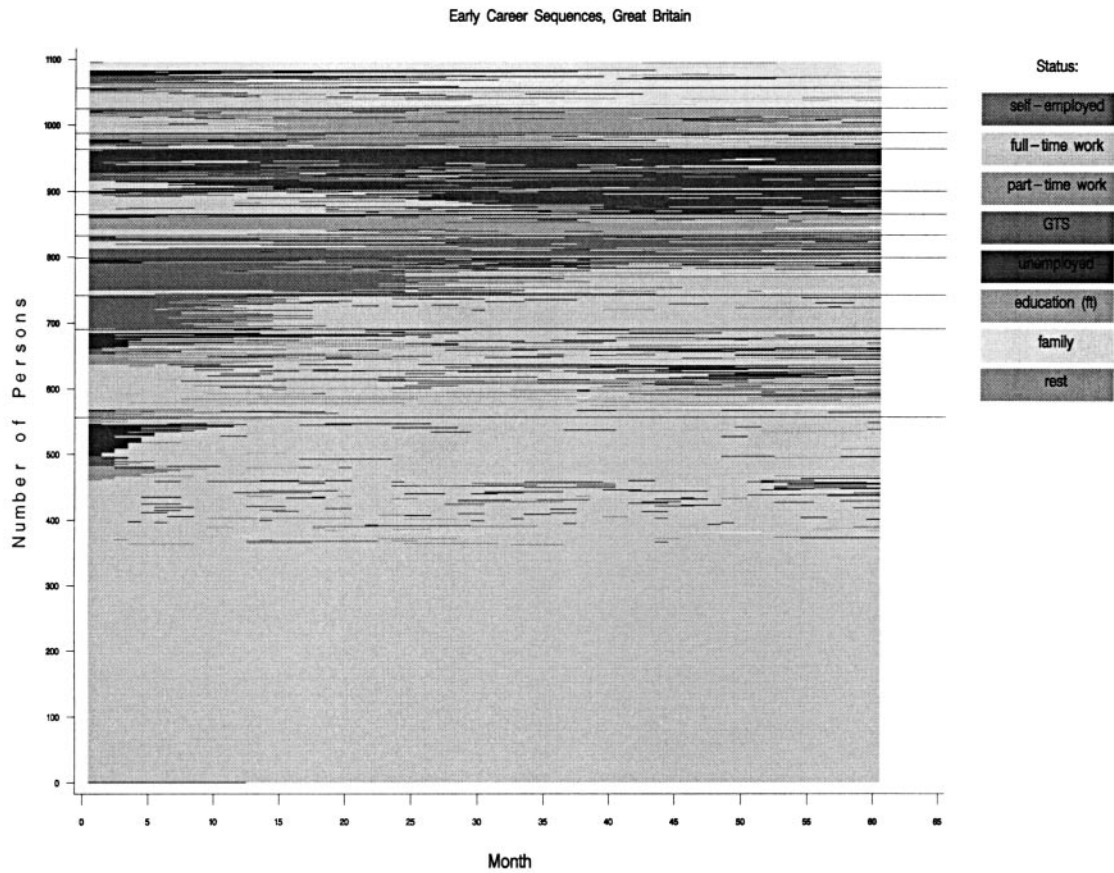
^aLabour-market entrants in this period.

Source: BHPS and SOEP, labour market entrants (first 5 years after leaving the ETS). Composition of the Clusters.

first remaining for about one year, the second about two years on average (see Table 4). These two groups clearly reflect the extension of the training time assigned to GTS over time: in the main the older respondents remain only briefly in this status. Government training schemes in GB are an instrument aiming to help especially less-educated young people to establish themselves in employment. Consequently we find higher proportions of less-educated young people in this group. Also the high proportion of young people with vocational training (about 60–67 per cent) corresponds to expectations, given that GTS is rewarded as a basic vocational qualification (CASMIN 1c). But this also indicates the relatively high proportion of GTS leavers without certificates, which is combined with prolonged training periods, decreasing over the two entry cohorts. The results seem to indicate that the instrument of GTS has succeeded in integrating young people into work: in the sample most of the respondents taking part in this scheme, independent of duration, show a smooth transition

into work afterwards, with only a few exceptions located in other clusters ending up in employment or family care. This, in a sense, supports the argument about the importance of work-based training, independent of standardized training content. Although it is not possible to assess the efficiency of this measure with this approach.

In common with other research (Lohmann, Luber and Muller 1999), the results show that self-employment is not very common in early careers. While for GB one cluster dominated by self-employment occurs, we obtain two groups for Germany that differ with regard to the time spent within this status. Besides the very few individuals entering self-employment directly after school and remaining within it, in principle all groups share the same patterns, with self-employment located at the end of these five years after a period in full-time dependent work. The higher proportion of people with spells of self-employment in Germany is somewhat surprising, as it clearly contradicts other results. However, it seems plausible that this

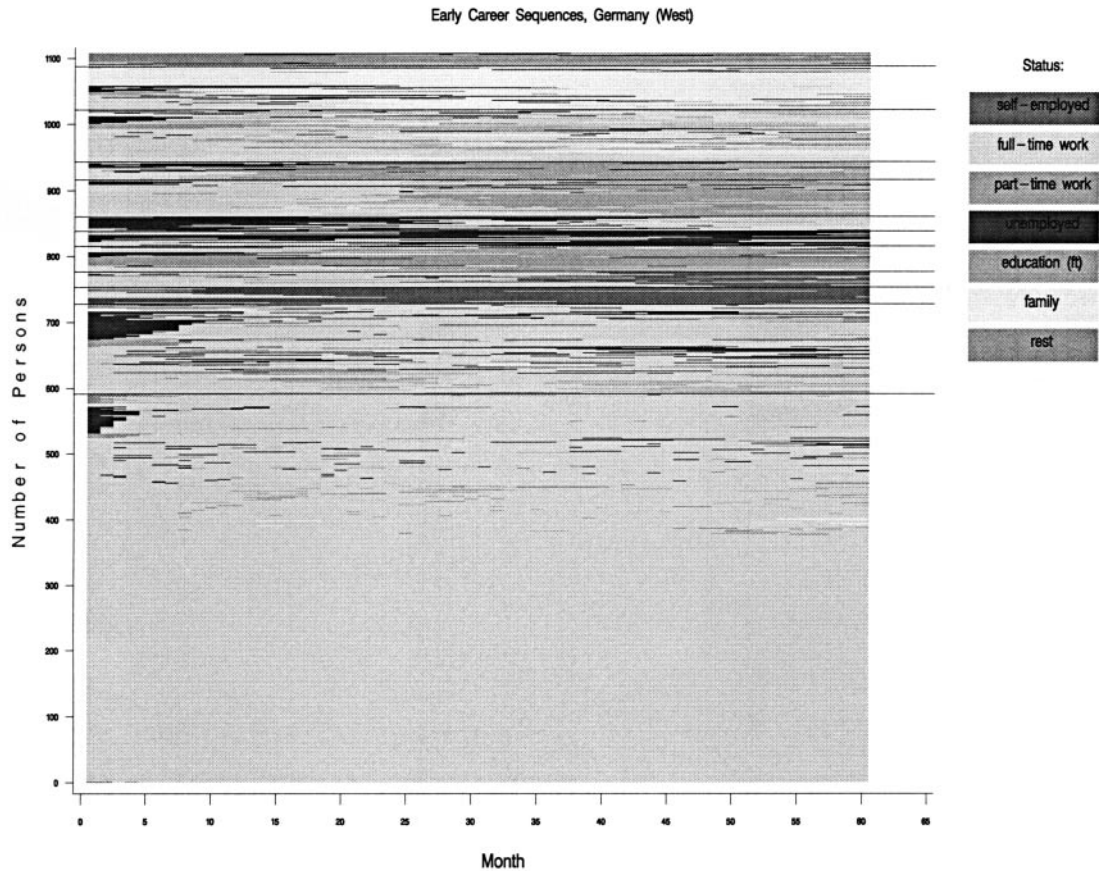


Source: BHPS, labour market entrants (first 5 years after leaving the ETS), individual career sequences. A coloured version of the graph can be found under: <http://www.mzes.uni-mannheim.de/publications/wp/wp-7-plot.pdf>

Figure 4a. *Early career sequences sorted according to their cluster membership, Great Britain*

result is due to the fact that information about self-employment is only available in a yearly grid for Germany and therefore, although corrections are made (see Appendix), the proportion of self-employment is overestimated. For this reason these results should be treated with caution. Due to sample construction overestimation of this status at the same time means underestimation of full-time dependent work in Germany.²⁵ In both countries the self-employed groups are mainly composed of males, but this gender difference is much more pronounced in Germany, indicating that self-employment in general is very unusual for women

in that country. For GB this pattern of self-employment is losing importance over the two entry cohorts, which seems surprising given the growing overall proportion of self-employment in Britain, while in Germany it became more common among the 'younger' entry cohort, who left full-time education after 1989. Self-employment in Germany is strongly dependent on educational qualifications based on the formal regulation of access. Vocationally trained individuals are over-represented among those starting self-employment after initial full-time dependent work, while it is the more highly educated respondents who start immediately with



Sources: SOEP, labour market entrants (first 5 years after leaving the ETS), individual career sequences. A coloured version of the graph can be found under: <http://www.mzes.uni-mannheim.de/publications/wp/wp-7-plot.pdf>

Figure 4b. *Early career sequences sorted according to their cluster membership, Germany*

self-employment. Given the different educational distributions of the two self-employed groups it seems plausible to think of them as two different types of self-employment. This contrasts with the British pattern, where education of this group is equivalent to the mean, indicating different structures of self-employment in the two countries. The same probably applies for the part-time work groups. In GB part-time work is more common among persons without any vocational training, supporting the idea of part-time work being largely combined with precarious employment (Schör,

1987; Meyer, 1997), whereas no over-representation of educational groups can be found in Germany.

Career courses predominantly consisting of part-time work participation emerge as a separate cluster in both countries, roughly equal in size to about 3–4 percent of the sample. These groups consist of status homogeneous careers as well as careers ending up in part-time work after a period of either unemployment, family care, or full-time work participation (see Figures 4a and 4b). The proportion of sequences starting with part-time work and subsequent transition into full-time paid work is quite low and located

within other clusters. The most important difference between the two countries with regard to this part-time working pattern is the composition by gender: while for Germany part-time work is clearly a female-dominated phenomenon, a considerable proportion of males are included in this group in GB, with growing importance over time. This results in a sharp overall increase of this group. Findings are in line with the expected higher change in GB going along with the greater degree of labour market flexibilization; male part-time work can be understood as an indicator of labour market flexibility.

Aside from these groups with more or less successful transitions into work, a considerable proportion of people follows patterns showing remarkable problems in entering work: namely, those with spells of unemployment in their early careers. Looking at involuntary labour market absence in the early career, differences among the two countries are most apparent. First, the higher portion of careers showing spells of unemployment in GB is striking, but we also find differences with regard to the ordering and the duration of this status. Considering for each country only the two emerging clusters dominated by unemployment, in GB they are more than twice as large as in Germany, covering about 9 per cent of the ETS-leaving cohort in contrast to only about 4 per cent. In addition to these two clusters, patterns containing spells of unemployment of shorter duration also occur, although located in other clusters. It is plausible to depict theoretically different unemployment patterns with regard to ordering and location of this status within the sequence. First, it is possible to distinguish those with only short periods of unemployment. This pattern exists in both countries, some of them, depending on the length of the period, grouped within the full-time work cluster and some within the more heterogeneous cluster of full-time working people with additional spells in other statuses.

Second, and at the other extreme, the pattern of long-term unemployment is apparent: in both countries a pattern of ongoing unemployment over the whole five years under investigation can be seen, though differing sharply in size. With regard to its frequency this group is of almost no importance in Germany, but considerably more common in GB. In Germany these persons are grouped together with

those who enter unemployment after a period within full-time work. This third configuration emerges as a separate cluster in GB, whereas in Germany the second unemployment-dominated group consists of those who succeeded in finding work after a considerable period of unemployment, as a fourth configuration. Labour market exclusion is strongly structured by gender, with gender differences depending in different ways on the patterns of unemployment in the two countries. The phenomenon of either having problems in entering work or even failing to enter is one that is definitely more common among men and clearly increasing over time in GB: men and the younger entry cohort are over-represented within these groups. The picture for Germany is not as clear-cut. We find shrinking proportions of the male-dominated pattern of initial unemployment at labour market entry over the two cohorts, while it is mainly women who are continuously excluded from the labour market. Although these findings should not be overestimated because of the small numbers they are based on, it seems as if in Germany women are at higher risk of being excluded from employment, whereas British women seem to find ways of escaping long-term unemployment. This second group of the long-term unemployed is also increasing slightly in size over time. Looking at the more unstable pattern which shows additional spells of full-time work, neither gender differences nor considerable change over time can be found.

Turning to the educational achievements of those individuals who follow the unemployment-dominated trajectories, findings are clearly in line with already well-established results (Brauns, Gangl, and Scherer, 1999 among others): it is mainly the less-educated as well as those without any vocational qualification who are affected by unemployment, disregarding duration or ordering of this status. In particular the group of long-term unemployed in Germany is strongly dependent on educational achievements, both academic and vocational. Correspondingly vocational training as well as possession of higher education facilitates access to more or less stable and ongoing full-time work, with the level of education being of greater importance in GB.

We now turn to those groups of individuals who leave the workforce for a considerable period in

their early career.²⁶ In both countries a substantial share of individuals (of about 3 to 8 per cent) return to education after entering the workforce, most of them after having entered full-time work.²⁷ This phenomenon is clearly much more common among the more recent entrants, showing the increasing importance of further education over time. At first sight the larger proportion of returners to education in Germany seems astonishing, considering the overall higher proportion of further education given in GB. As this training in GB is mostly part-time courses, it becomes clear that it is not taken into account in this approach, drawing on respondents' main status only, which means disregarding additional statuses. Returners to education are, at least to a large extent, bundled together in one group for GB, all following the pattern of a rather short period within full-time dependent work of about one year and subsequently remaining within education for the rest of the period under study. The same pattern is found in Germany but the higher proportion of people making up a group with longer periods of previous full-time work. With regard to the patterns described first, this is probably a typical waiting loop situation: young people enter work as a transition stage to the next period of education (and due to sample construction this formation is notoriously small in Germany²⁸). Few individuals choose education as a way to escape unemployment. Return to education goes along with higher educational achievements, but not with vocational training. Although focusing on the composition of this group with regard to educational achievements is problematic, given the static measure applied here, findings provide a first rough picture. It suggests that it is primarily tertiary-level training participants who are included in this pattern.

While participation in further education does not differ by gender, it is only women who leave or never take up paid work for reasons of private family care activities (with the exception of one man in Germany). In principle the same patterns occur in both countries, but proportions vary clearly. The two family care groups emerging in each country taken together are about double the size in Germany. More precisely, in Germany one group contains those directly entering this status and those with short periods in work before

leaving the workforce. This direct entry and early exit is more common in Germany, which fits into the picture of more traditional roles of women in that country.²⁹ A second rather heterogeneous group consists of those with shorter spells in family care that are not clearly structured in location.³⁰ This group also contains males, since not all persons in this group actually show spells of domestic work. The grouping of the patterns in GB differs. One cluster contains those in ongoing family care and comprises individuals starting with spells of either unemployment, GTS, or part-time work. In contrast to Germany the second is rather clearly structured, containing females leaving full-time work after about two and a half years. We could speculate that those with preceding unemployment and probably those with GTS spells choose family care or maternity as possible alternatives because of their inability to become established within the labour market. But as we know nothing about their personal reasons, this cannot be investigated. The often reported high proportion of female labour market withdrawal in order to avoid unemployment in GB cannot be found through this approach, which is probably due to the fact that it is early careers that are the focus. Family care decreases in importance for GB, indicating that females either leave work less often or later in their career for that reason,³¹ but has remained more or less stable in Germany. A much higher and surprisingly stable proportion of women in Germany leave the workforce early in their career to become a housewife or mother. Additionally women seem to leave work for family or maternity reasons regardless of their educational achievement. In contrast the choice of family care in GB is strongly structured by educational achievement: women with higher education or vocational training leave the workforce less often for that reason and instead remain in work. Especially among those who leave the workforce early and subsequently spend long periods in family care activities, a lower level of education and the lack of vocational qualifications is common.

In addition to the clusters described above, for both countries one cluster composed mainly of those with spells in the 'rest' category emerges. On average they spend most of their time in either

military service (for GB only, see appendix), retirement, or a not further specified 'other' category. In GB this group also has some periods in full-time work after preceding spells in other statuses such as unemployment, education, GTS, or the 'rest' category and is primarily composed of men. In Germany this group, a growing proportion over time, is mainly composed of those in the 'other' category including some with additional spells of education.

Summary and Discussion

This paper's major aim was to apply a new method, sequence analysis, to the topic of interest: the investigation of entry patterns into work and their variation between different institutional settings. More precisely, the question of whether entry patterns can be identified empirically and how they differ between different institutional contexts, namely between West Germany and Great Britain, was posed. Differences between the country-specific patterns of entering employment are expected to occur mainly in three ways, first as different types of emerging patterns, second with regard to their importance in terms of frequency, and third in their composition, including gender differences, changes over time, and the predominance of specific educational credits. In a nutshell, with the applied method we are able to identify typical early career trajectories, revealing shared as well as country-specific patterns in the two economies. The investigation of the early career patterns focuses on the first five years after leaving the education and training system, drawing on monthly employment status information. Through the distinguished statuses, periods in employment, unemployment, and periods of non-participation in the workforce have been covered. For both countries patterns of continuous full-time work participation, which make up the highest proportion in both countries, are found, patterns indicating problems entering the labour market, groups clearly not successful in making the transition into work who spend most of their time within unemployment, as well as patterns of self-employment, part-time work, and clusters leaving the labour force, if they ever entered it at all, for further education or family reasons. For

GB separate clusters containing GTS participants emerge, which of course have no counterpart in Germany. In sum, using the chosen approach we find country-specific as well as similar typical career pathways in the two countries. Most of the emerging clusters have a counterpart within the other country. Differences occur mainly with regard to the overall frequency – the importance of the specific trajectories in the two countries – and their composition with regard to gender, loss or gains in importance over time, and to some extent the educational achievements.

The introduction has outlined some assumptions about the ways in which specific institutional and structural surroundings shape the process of entering employment, resulting in different entry patterns in Great Britain and West Germany. Of special importance for this process are features of the ETS and the labour market structure. In Germany transition into work is expected to be smooth and well regulated, whereas the process is supposed to be characterized by much higher instability in GB. Although in both countries the large majority of individuals follow patterns of predominantly full-time work participation, support for the outlined hypotheses emerges with respect to labour market exclusion as well as to unstable career patterns. Differences between the two countries become most apparent with regard to the incidence of unemployment: the proportion of unemployment-dominated careers is more than twice as high in GB as in West Germany. A considerable proportion of those in GB, nearly 10 per cent, fail to enter the labour market within this first five years. In contrast this group is of marginal importance in West Germany, where even the unemployment-dominated pattern shows more or less successful transition into work after a period of unemployment. In addition to these unemployment clusters, patterns of much higher heterogeneity indicating problems entering employment emerge for both countries. They are quite equal in size but differ in their heterogeneity, with patterns being more unstructured in GB. Taken together, this supports the idea of more stepwise and problematic entry processes in GB characterized by greater turbulence, whereas transition is more clearly structured in West Germany, which also leads to the predominance of stable patterns.

As alleged above, change over time in the way labour market entry takes place is supposed to be more prevalent in GB and hardly existent in West Germany. Besides the fact that the GB labour market is usually referred to as being much more flexible than especially the German one, this flexibility has also been increasing over recent decades. For this reason a higher and growing proportion of non-standard working conditions is expected for GB. Evidence for this view can be found in the growing proportion of male part-time work participation, indicating the erosion of traditional working conditions for men accompanying higher flexibility. As patterns of labour market exclusion also grow in importance over time, it seems obvious to interpret these changes as a sign of changing mechanisms of accessing the labour market and the matching process to stable jobs as well. These findings confirm the idea that entry takes longer and is becoming increasingly problematic in GB: it takes more time to enter work and become established after the first try, and patterns are becoming increasingly unstable. However, to fully address the issue of change over time, a broader time span and more distinct entry cohorts are needed.

Labour market structures have no necessary implications for the amount of gender segregation of labour but often go along with it. Work participation in West Germany is highly structured along gender lines. This is true for labour force participation as well as for specific occupational choices. Although gender segregation cannot be explored in detail here, evidence can be found with regard to part-time work participation and family care activities. If women in West Germany participate in paid work at all, they follow part-time working careers to a much higher degree than their male counterparts. So, while part-time work in West Germany is therefore a clearly female-dominated phenomenon, it is of much higher importance among British men, sharply growing in importance over time. A considerable proportion of women in West Germany, however, follow the 'traditional pattern' of leaving the work force for family care activities. These patterns remain remarkably stable over the two entry cohorts under investigation, whereas they lose importance for British women.

Through the application of this rather new methodological approach to handling career course information we can extensively enlarge the perspective on labour market entry processes and the early career patterns typically followed. Sequence analysis allows for the handling of whole career trajectory information, considering the serial succession of different statuses, their duration and ordering, instead of only single events or single time points. That this is not just a methodological issue but also of substantial importance becomes clear if we think of labour market entry as a process evolving over time. This process can only be fully captured by this kind of holistic approach. Thus, this approach opens an important perspective on the investigation of career patterns and consequently on the comparison of different institutional arrangements.

So, the question about the underlying mechanism producing different kind of work career patterns can now be addressed in a new and very promising way. In addition, sequence analysis also enables the investigation of typical career patterns and their diversity over different institutional settings, based on a much broader set of information, both with regard to sample size and career information. This has important implications for the development of typologies of trajectories. Such typologies are especially important for the comparison of entire nations, and the more differentiated the typologies are the more information we can hope to extract about the impact of the institutional settings.

In this paper I have followed a more or less 'empirical approach' to investigate typical early career patterns and their differences between the two countries. Although a 'theoretical' classification of the different patterns would have easily been possible with the limited number of statuses taken into account here, we soon reach the limits of that by expanding the status information considered. Focusing on employment status information makes classifying and interpreting the emerging typical career patterns as equivalent in the two nations a straightforward undertaking. However, it still is an open question to what extent the phenomena behind these statuses can be considered comparable or significantly different between the two economies. This is obvious with regard to full-time employment, which indeed covers a lot of heterogeneity,

but also if we look at self-employment or part-time work. In fact, there may be different situations people typically face. Focusing on the composition of the 'typical patterns' allows us to assess these potential differences to some extent. Thus, the chosen perspective on additional covariates was basically one of further qualifying the emerging patterns instead of explaining them. Therefore we focused more on the composition of the groups. With this view we give credit to the fact that the emerging patterns are not totally comparable but even further emphasize their differences. These differences for instance became obvious with regard to the part-time working group. The predominance of males versus females can be taken as an indicator of different types of employment. Absence of vocational training in this group in GB may also support that view.

A different perspective would be to concentrate on the factors affecting the pathways to be followed. For instance the country differences in educational systems outlined in the introduction and their specific link to occupational achievements suggest the existence of considerable differences in the way education affects access to specific job positions and subsequent occupational careers. By now it has become clear that education is a central dimension influencing the career pattern followed. However, the question about the impact of educational achievement on the early career pattern followed within the different national contexts and how the process of converting educational achievements into occupational positions takes place in these structures must be answered in future research. With this perspective we can then leave the more or less descriptive approach we have taken with this method up to now and return to a more explanatory perspective.

Although even this rather rough categorization of the employment status variable reveals interesting insights into the typical patterns of labour market entry and the institutional factors affecting them, it is not possible to fully investigate this process and its determinants. Therefore a further subdivision of the statuses studied is needed. With a more differentiated view of occupational positions and further job characteristics it will then be possible to assess the underlying matching processes in more detail.

Notes

1. The analysis of Germany is confined to West Germany, given the specific situation in the former GDR which would require separate analysis. As the British Household Panel Survey (BHPS) contains no data about Northern Ireland empirical analyses do not cover the entire UK, but are restricted to Great Britain.
2. These reforms are primarily aimed at countering the critique of an under-skilled workforce in the UK and reducing rising youth unemployment rates.
3. This assessment will probably change (again) following the recent development of a 'new vocationalism'.
4. Different definitions of the transition period have to be distinguished. Often education after compulsory schooling is conceived as part of the transition period. In this case the transition is defined as starting at a specific age. Others include only the period after leaving education, which may be somewhat problematic for the specific case of the German apprenticeship system. As this paper focuses on entry processes into work and not so much on the various forms of transitions leading to that entry, the latter definition is used.
5. Through the introduction of insertion/deletion transformation it becomes possible to detect sequences that are pretty equal even when they are rather staggered and, for instance, differ in status at every single time point.
6. The problem here is even more complex as we also have to deal with two different datasets, one for Germany and one for Great Britain.
7. Measurement of the homogeneity is by the variance criteria. Ward has the tendency to produce equally sized groups and has difficulty in detecting very small groups.
8. Concerning the BHPS (Taylor, 1996), I draw on the combined datasets, which include long-term retrospective as well as the direct information from the panel waves (Halpin, 1997).
9. Reduced numbers in the analysed models are due either to missing data or, in case of the sequence analysis, to gaps in the sequence; distances are not computed in that case.

10. The exclusion of apprenticeship seems reasonable on several grounds: one has to consider apprentices as having a special status which gives them a very stable position and makes their condition not comparable to other 'normal working conditions'. Another point is that because of the relatively long duration and the large proportion in apprenticeship training in Germany, most of these five years under investigation would mainly consist of periods of apprenticeship. It could be useful to treat this institutionalized 'transition phase' separately.
11. It can be shown that young men in Germany undertake their military service in almost all cases directly after periods in education, either between two education spells or before labour market entry, depending on their level of education.
12. It does, however, have the disadvantage of being relatively crude, hiding considerable degrees of mobility between different employment statuses. Therefore a further subdivision of employment positions could be a topic for future analysis.
13. Government training schemes cover Youth Training or Youth Training Schemes, for instance.
14. Consideration of this extra status leads – *qua* definition – to more possible mobility in the British case. It seems reasonably justified, however, taking into account the specific national situations.
15. With this distinction we do not include business cycles. Unemployment rates peaked in the mid-1980s, fell in the late 1980s and rose again with the beginning of the 1990s. So what we get by this approach is something like 'entrants into a relaxing labour market' for the first entry cohort and for the second one 'entrants into an increasingly competitive one' (OECD, 1995).
16. Education is measured as the highest education achieved in 1991. As there is no information about education available for the British respondents before then, it is the time point closest to leaving the educational system. Only where there is no information at this point in time is the status of later years considered.
17. The sequence alignments are calculated using TDA, a programme developed by Rohwer (1996).
18. This, however, is not due to higher women's part-time work participation, as we shall see later.
19. Direct comparison of distances is not recommended given the one additional status in GB, which leads to an enlargement of distances. It is also important to note the fact that we only have women in family care activities, which means a larger state-space leading to greater distances. This however is also a substantive result.
20. This is a very crude approximation and to disentangle both processes more sophisticated modelling would be needed.
21. This, in some sense, is surprising, given the one additional status in GB. Closer inspection revealed no further meaningful subdivision when enlarging the number of groups. Some clusters, however, are more heterogeneous in GB owing to this additional status. The 'big clusters' appear very early within the clustering procedure and remain relatively stable, indicating that they are quite homogeneous. So what happens at the end of the clustering procedure is the merging of relatively heterogeneous, small subgroups. It may also be of interest that we do not obtain the same cluster solution by clustering the data based on the amount of time spent in the 7 (FRG) or 8 (GB) statuses. This shows the importance of the additional information concerning the ordering of events taken into account by OMA.
22. An alternative strategy would consist of running cluster analysis on the combined datasets and splitting up the groups according to the country information afterwards. However, this would, in my view, lead to even more problems in understanding what happens during this procedure.
23. Remember that only the first 5 years are under investigation. Therefore a mean duration of about 60 months would mean that almost all respondents spend their whole early career within this single status.
24. More precisely, to distinguish the level of education, every qualification up to 2b and including, was grouped within the 'low level qualification' and those equal or above 2a in the 'high level'

- category. To construct the vocational training information 1c, 2a, 2cvoc, and 3ab are grouped together.
25. Periods of self-employment were only filled in if the main status was full-time work.
 26. In principle the ILO definition of workforce is adopted in this paper, with slight modifications, however: individuals in full-time education are not regarded as belonging to the workforce.
 27. By definition there are no respondents starting with spells of education. Return to education in the British data is defined as respondents' own assessment of their main status, whereas multiple states in Germany are allowed for so that a hierarchy of states has been introduced. This potentially leads to an overestimation of periods in education (see Appendix).
 28. The chosen sample selection in Germany is supposed to avoid including young people in their waiting loops between two spells of education (see the Appendix for details of the sample construction).
 29. Including the former East Germany would change this picture considerably.
 30. It should be mentioned that these patterns differ if substitution costs in calculating distances are weighted. As the introduced weighting gives greater emphasis to the occurrence of specific states, the pattern is more clearly structured, especially for Germany. In the case of weighted distance measures, two groups emerge, the first containing those with long periods of family care, the second with short periods. Looking at the patterns in this way the higher proportion of females in this status becomes more apparent.
 31. It is possible that these differences are at least to some extent due to the lower age of females in GB in this school-leaver sample. The education and training system in general is left earlier in GB so that the investigated sample in GB is assumed to be younger. In fact people are on average about four years younger in the British sample.

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Appendix

The Sample

The chosen approach covers the first five years after leaving full-time education. Although this definition is rather precise, it leads to some practical problems concerning the sample selection. While the long-term-retrospective part of the BHPS exactly reports the careers after leaving education the first time, the situation is not that easy for the information coming from the Panel Waves in GB and Germany. To make sure that school leavers are included, only respondents with first spells in education are selected, as we otherwise do not know in which stage in their career people take part in the survey. This strict definition is enlarged by including young people (under 26) with spells of education other than their first. In this way (labour market) activities before finishing education are allowed. A considerable proportion of young people do not have their education spells in succession but with some other activities between. In principle they are included in the sample after their last education spell, with some exceptions. If there is a considerable time period between two education spells, respondents join the sample after the first period. Coming from apprenticeship, a half year of other

activities is allowed for, whereas from school or university the threshold is one-and-a-half years. With this procedure ‘waiting loops’ are taken into consideration. Young people often do not enter higher education directly but increasingly experience some kind of ‘waiting loops’: e.g. young people waiting for a place at university, apprentices deciding to attend further education, or simply young people not knowing what to choose. Within this period we cannot think of them as really entering the labour market. In addition, for Germany sequences are only selected after military service.

The status information as defined is directly provided within the BHPS, while the status variable within the SOEP has to be enlarged by including information about self-employment. Self-employment is unfortunately only available within a yearly grid. It was taken into account as follows: if a person indicates a time period of full-time work and she also reports being self-employed within the yearly grid, this spell up to the end of the year is replaced by self-employment. But with this procedure we have to assume an overestimation of the time spent in self-employment.

With the SOEP data we have to deal with more than one piece of status information per time point. Therefore it was necessary to introduce a hierarchy of statuses which was done as follows: full-time education, full-time work, part-time work, unemployed, maternity leave and homemaker, followed by a ‘rest’ category. Thus, for a person reporting both full-time education and part-time work for the same period, priority is given to full-time education and any other information about further statuses is ignored. This was necessary for the sample selection but may lead to an overestimation of education. Additionally there are a large number of gaps between the serial episodes. In this case the missing information was filled in by either previous or subsequent reported spells taking into account the information provided about the censoring of the statuses.

Educational achievements are measured as the highest educational qualification people have in 1991. This was the best possible approximation to the educational achievements at labour market entry as no information about previous time points is available for Britain.

Table A1. *The CASMIN scale of educational qualifications*

Qualification	Description
1ab	This is the social minimum of education. Namely, the minimal level that individuals are expected to have obtained in a society. It generally corresponds to the level of compulsory education
1c voc	Basic vocational training above and beyond compulsory schooling
2b	Academic or general tracks at the secondary intermediate level
2a voc	Advanced vocational training or secondary programmes in which general intermediate schooling is combined by vocational training
2c	Full maturity certificates (e.g. the Abitur, Matriculation, Baccalauréat, A-levels)
2c voc	Full maturity certificates including vocationally-specific schooling or training (e.g. Baccalauréat de technicien)
3a	Lower-level tertiary degrees, generally of shorter duration and with a vocational orientation (e.g. technical college diplomas, social worker or non-university teaching certificates)
3b	The completion of a traditional, academically-oriented university education

Source: Adapted from Brauns and Steinmann 1999 and Müller and Shavit 1998.

Calculation of the Matching Costs

Calculation of the dissimilarities in the compared sequences is based on two distinct transformations for which different costs can be assigned: substitution and insertion/deletion (abbreviated as 'indel' transformation). Introducing indel costs instead of reducing the admitted transformations on substitution is important to be able to take into account the ordering of specific statuses in the sequence. This becomes obvious if we think of two sequences with the same statuses but in a different order. Two years of full-time work and two in family care is obviously more similar to the first two years within family care and the last two in full-

time work than, for example, continuous part-time work. In fact, these sequences are different in any time interval we consider. So – besides the fact that we can introduce different weighting – only allowed for substitution would mean calculating the same distance for both possible pairs. Admitting indel costs, however, makes it possible to cut the first two years of full-time work, move the last two years to the beginning, and add two years of full-time work at the end of the sequence to make it similar to the reference group, instead of changing the statuses of every single time interval. In this example it becomes clear how the assigned weights of substitutions and insertion/deletion depend on each other.

Table A2. *Substitution costs underlying the distance calculation of the sequences*

	Self-empl.	FTempl.	PTempl.	GTS	Unempl.	FTeduc.	Family	Other
Self-employed								
Full-time Employed	1							
Part-time Employed	1	1						
GTS ^a	1.1	1.1	1.1					
Un-employed	1.3	1.3	1.3	1.2				
Education (full-time)	1.4	1.4	1.4	1.3	1.3			
Family care	1.45	1.45	1.45	1.3	1.4	1.3		
Others	1	1	1	1	1	1	1	

^aGovernment Training Schemes (GTS) is not considered for Germany.

Note: Costs for deletion and insertion: InDel=0.75.

Setting the indel cost to equal half of the substitution cost is an inserting option as it makes the costs for two indel operations equal to one substitution (Rohwer, 1997). This approach introduces no weighting at all, neither with regard to indel or substitution nor to the different statuses. Non-weighted results presented in the paper are based on this calculation. Indel operations are each assigned a weight of 0.5 and substitutions one of 1.0. However, as it seems plausible to think of some statuses as being closer together than others, I decided to introduce some weighting of the substitution costs. Therefore a hierarchy of statuses has to be introduced. The underlying idea of the development of the applied cost structure is the usual division of (a) 'in labour force' composed of: employed/self-employed and unemployed, and (b) 'out of labour force'. These divisions constitute the main thresholds and weighting of the statuses is in some sense to be understood as nearness to the labour market.

Although this approach takes into account the ordering of statuses, it was not possible to assign different weights to the same transformations at different points in time: the location within the sequence is not relevant for calculating the costs. Rohwer and Trappe (1997), for instance, propose to overcome this shortcoming by calculating

'sequences for costs' for the sequences. But this is where we reach the limits of the method for the moment.

Closely connected to this point is the problem of treating sub-episodes as if they were independent. This is a problematic assumption with regard to career sequences, as the probability of a specific status is already a function of the preceding sequence. This becomes clear if we consider that most of the episodes in the sequences are of longer duration and therefore the probability of continuing with an already started status is higher than a change in status. This could be taken into account by attributing higher costs for change in status than for their continuation, which would provide a more dynamic view (Erzberger and Prein, 1997). At this point the potential of the method has to be elaborated further.

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